FUTURE BATTERY INDUSTRIES CRC

# Tools to Determine the Best Solution for Mining Vehicle Electrification

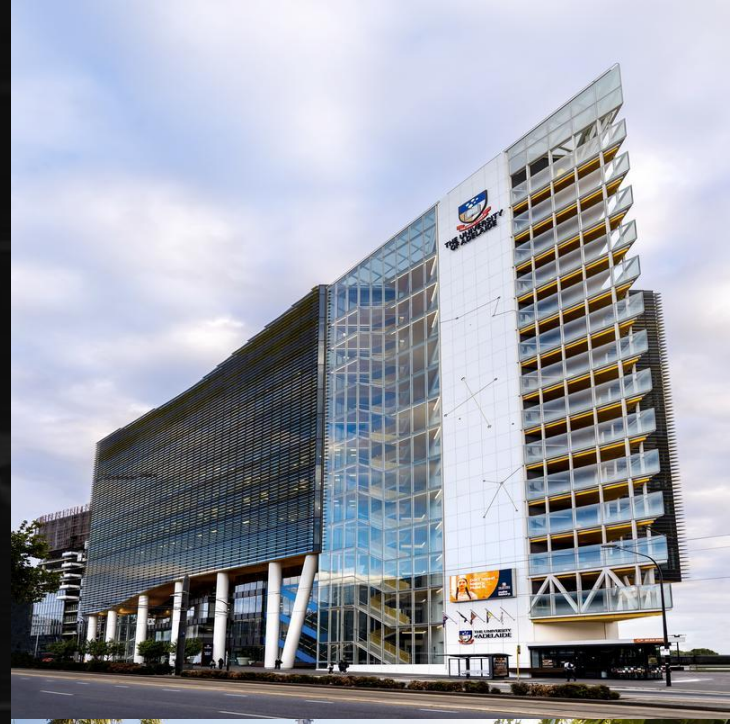
Dr Ali Pourmousavi
The University of Adelaide

THE UNIVERSITY of ADELAIDE

The MOVE Project team acknowledges and pays respect to the Kaurna People, the traditional custodians of the Adelaide Plains region, whose lands the MOVE Project is located.

We acknowledge the deep feelings of attachment and relationship of the Kaurna people to country, and we respect and value their past, present and ongoing connection to the land and cultural beliefs.

We also pay our respects to Aboriginal and Torres Strait Islander peoples across Australia.



# Contents


1. About the MOVE project
2. Background and motivations
3. Problem statement
4. Results of a simulation study
5. Our service

# 1. About the MOVE Project

# About MOVE project

| | |
|---|---|
| **PROJECT TITLE** | Assessment, design and operation of battery-supported electric mining vehicles and machinery |
| **ULTIMATE GOAL** | To provide the Australian mining industry with the tools and information needed to help transition their operations to using battery-supported electric vehicles and associated stationary machinery |
| **2 CASE STUDIES** | • Leinster Underground Mine, WA (BHP)<br>• Nova Operation, WA (IGO) |

# About MOVE project

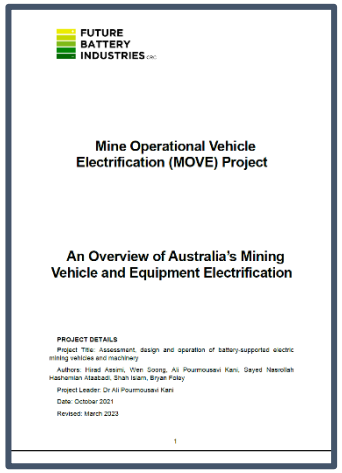
1. ENERGY system study and data collection

2. DEVELOPMENT of tools for EV fleets and associated infrastructure

3. DEVELOPMENT of the toolkit to design backbone energy infrastructure

4. IDENTIFYING Electric mine required skills and producing training materials

# About MOVE project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Mine Operational Vehicle Electrification (MOVE) Project

An Overview of Australia's Mining Vehicle and Equipment Electrification

PROJECT DETAILS

Project Title: Assessment, design and operation of battery-supported electric mining vehicles and machinery

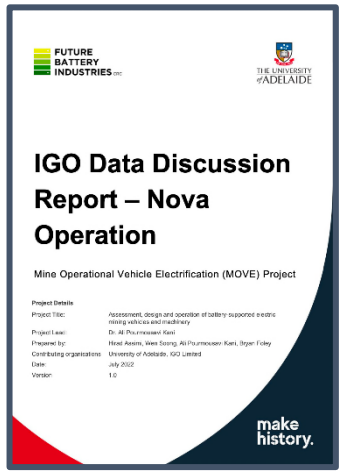
Authors: Hirad Assimi, Wen Soong, Ali Pourmousavi Kani, Sayed Nasrollah Hashemian Abadadi, Shah Islam, Bryan Foley

Project Leader: Dr Ali Pourmousavi Kani

Date: October 2021

Revised: March 2023

1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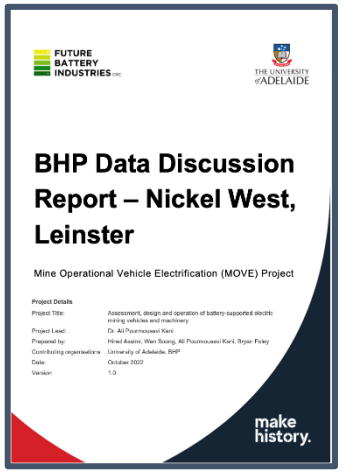
IGO Data Discussion Report – Nova Operation

Mine Operational Vehicle Electrification (MOVE) Project

Project Details

| | |
|---|---|
| Project Title: | Assessment, design and operation of battery-supported electric mining vehicles and machinery |
| Project Lead: | Dr. Ali Pourmousavi Kani |
| Prepared by: | Hirad Assimi, Wen Soong, Ali Pourmousavi Kani, Bryan Foley |
| Contributing organisations: | University of Adelaide, IGO Limited |
| Date: | July 2022 |
| Version: | 1.0 |

make history.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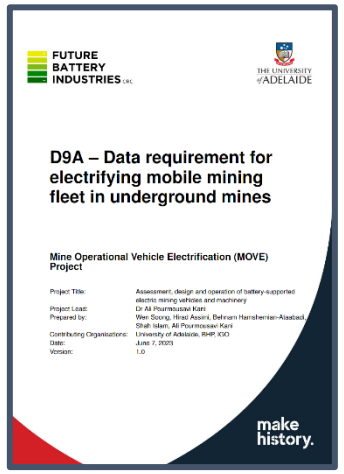
BHP Data Discussion Report – Nickel West, Leinster

Mine Operational Vehicle Electrification (MOVE) Project

Project Details

| | |
|---|---|
| Project Title: | Assessment, design and operation of battery-supported electric mining vehicles and machinery |
| Project Lead: | Dr. Ali Pourmousavi Kani |
| Prepared by: | Hirad Assimi, Wen Soong, Ali Pourmousavi Kani, Bryan Foley |
| Contributing organisations: | University of Adelaide, BHP |
| Date: | October 2022 |
| Version: | 1.0 |

make history.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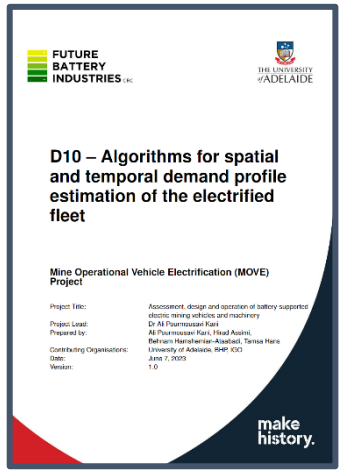
D9A – Data requirement for electrifying mobile mining fleet in underground mines

Mine Operational Vehicle Electrification (MOVE) Project

| | |
|---|---|
| Project Title: | Assessment, design and operation of battery-supported electric mining vehicles and machinery |
| Project Lead: | Dr Ali Pourmousavi Kani |
| Prepared by: | Wen Soong, Hirad Assimi, Behnam Hamsherian-Abadadi, Shah Islam, Ali Pourmousavi Kani |
| Contributing Organisations: | University of Adelaide, BHP, IGO |
| Date: | June 7, 2023 |
| Version: | 1.0 |

make history.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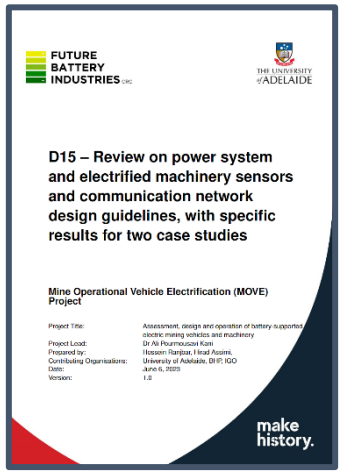
D10 – Algorithms for spatial and temporal demand profile estimation of the electrified fleet

Mine Operational Vehicle Electrification (MOVE) Project

| | |
|---|---|
| Project Title: | Assessment, design and operation of battery-supported electric mining vehicles and machinery |
| Project Lead: | Dr Ali Pourmousavi Kani |
| Prepared by: | Ali Pourmousavi Kani, Hirad Assimi, Behnam Hamsherian-Abadadi, Tamsa Hare |
| Contributing Organisations: | University of Adelaide, BHP, IGO |
| Date: | June 7, 2023 |
| Version: | 1.0 |

make history.


FUTURE BATTERY INDUSTRIES CRC

THE UNIVERSITY of ADELAIDE

D15 – Review on power system and electrified machinery sensors and communication network design guidelines, with specific results for two case studies

Mine Operational Vehicle Electrification (MOVE) Project

| | |
|---|---|
| Project Title: | Assessment, design and operation of battery-supported electric mining vehicles and machinery |
| Project Lead: | Dr Ali Pourmousavi Kani |
| Prepared by: | Hossein Ranjbar, Hirad Assimi, |
| Contributing Organisations: | University of Adelaide, BHP, IGO |
| Date: | June 6, 2023 |
| Version: | 1.0 |

make history.

# About MOVE project

Australian Government
IP Australia

Certificate of Grant
Standard Patent

The Commissioner of Patents hereby certifies that Patent 2013200300 has been granted under Section 61 of the *Patents Act 1990* to:

**Applicant Name**
Applicant Address
**Applicant Name**
Applicant Address

For the Standard Patent titled:
**Title**
Invented by
**Inventor name**
**Inventor name**

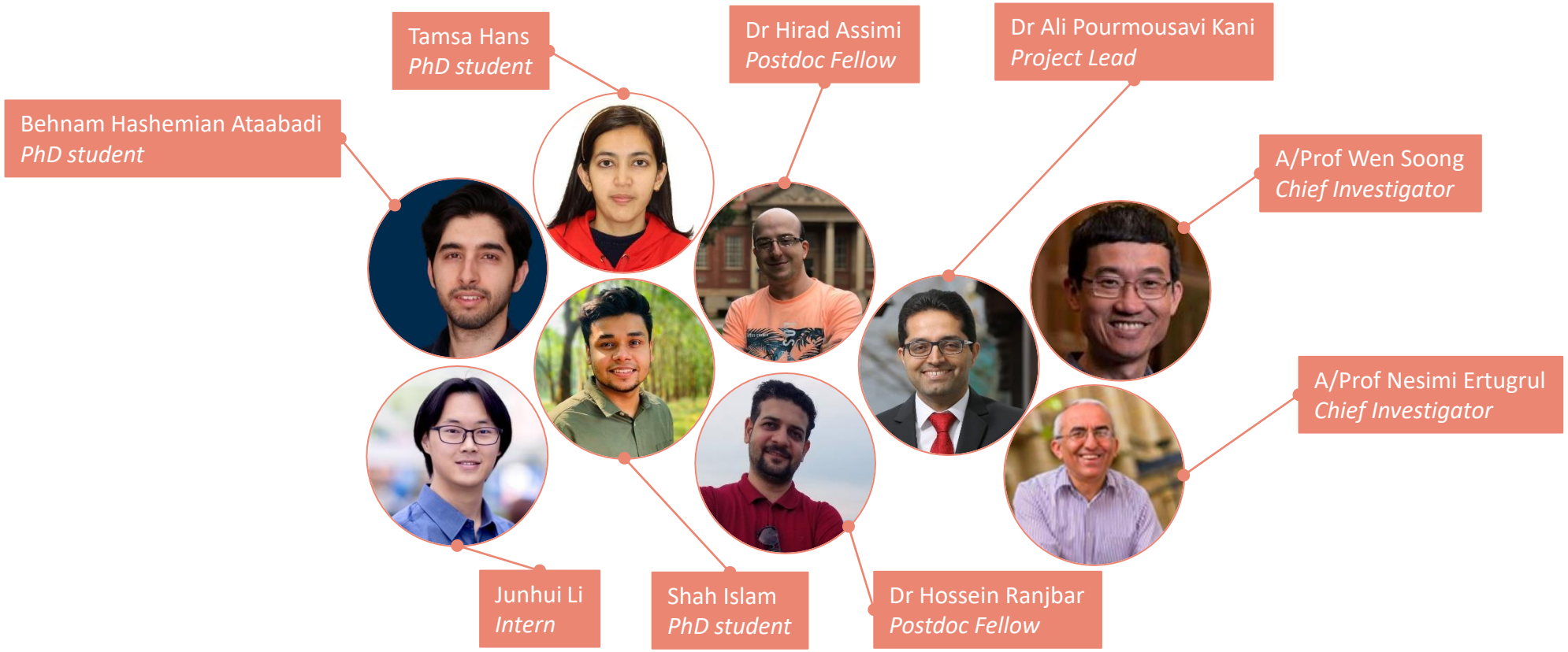
As specified under Section 67 of the *Patents Act 1990*, this Standard Patent is granted for a term of 20 years, commencing on <date of the patent>.
The term is subject to the payment of renewal fees

**Dated this 3rd day of June 2014**

Charging infrastructure sizing, technology selection and onboard battery sizing for electric haul trucks in underground mining application

Towards Underground Mobile Fleet Electrification
*Three Essential Steps to Make a Real Change*

Industrial IoT in Mine Electrification:
*Necessity or Luxury?*




A team of 3 academic staff members, 2 postdocs, 3 PhD students and an undergrad intern

School of Electrical & Mechanical Engineering
The University of Adelaide

# 2. Background and Motivations

# Background and Motivations

1. Australian mining industry consumes about 14% of the total energy usage in the country
2. Mining industry is the second largest contributor to carbon emissions in Australia
3. The fastest growing sector in GHG emissions for the past 30 years
4. About 30-50% of diesel fuel used at mine sites is for transportation
   15-30% of operating cost for material handling, heating, ventilation & processing
5. A report by McKinsey & Company: approximately 56% of emissions from diesel usage in an open-pit iron ore mine in Australia result from hauling.

# Background and Motivations

1

## Improve Mining Economics

- Reduction in Operating Energy Costs
- Reduction of Operating Maintenance Costs
- Reduction in Capital and Operating Ventilation Costs
- Reduction in Total Operational Expenditure (OPEX)
- Pathways to Mining Vehicle Optimisation and Automation
- Improvements in Lower Grade Ore Extraction Economics

2

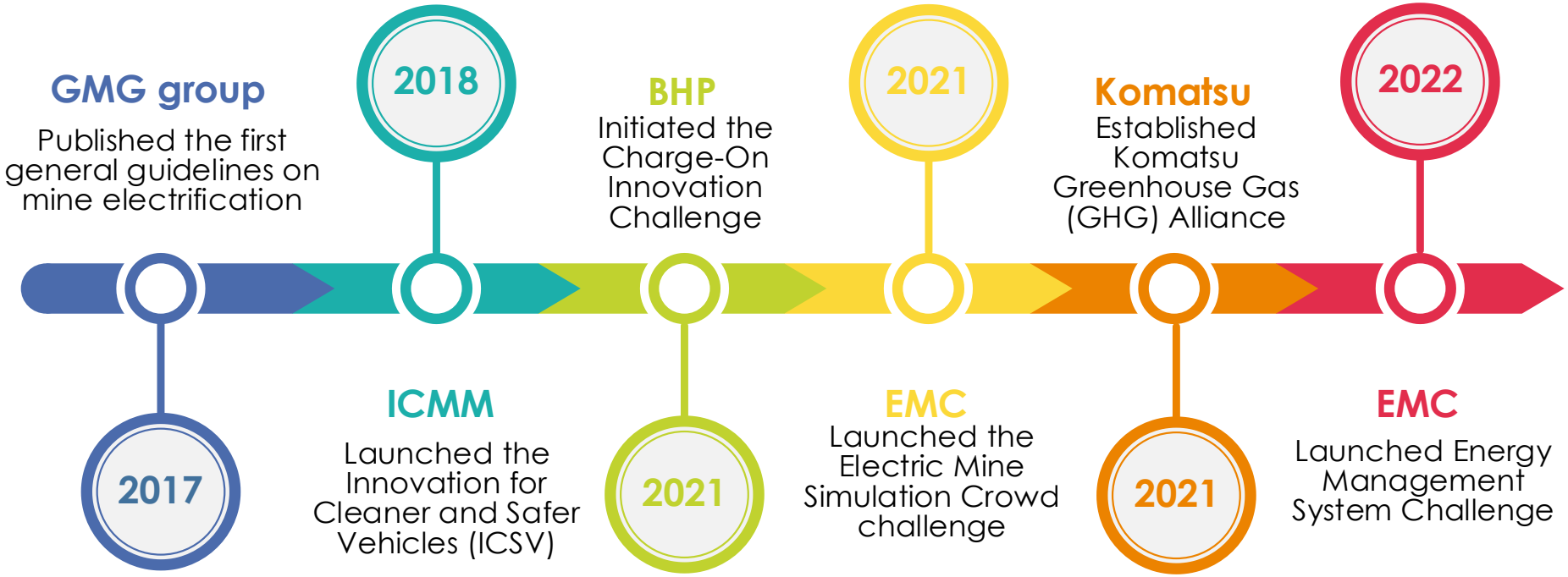
## GHG Emissions Reduction from Mining

- Greenhouse Gas Emission Targets
- Stakeholder Interests
- Society

3

## Improve Mining Worker Safety

# Background and Motivations



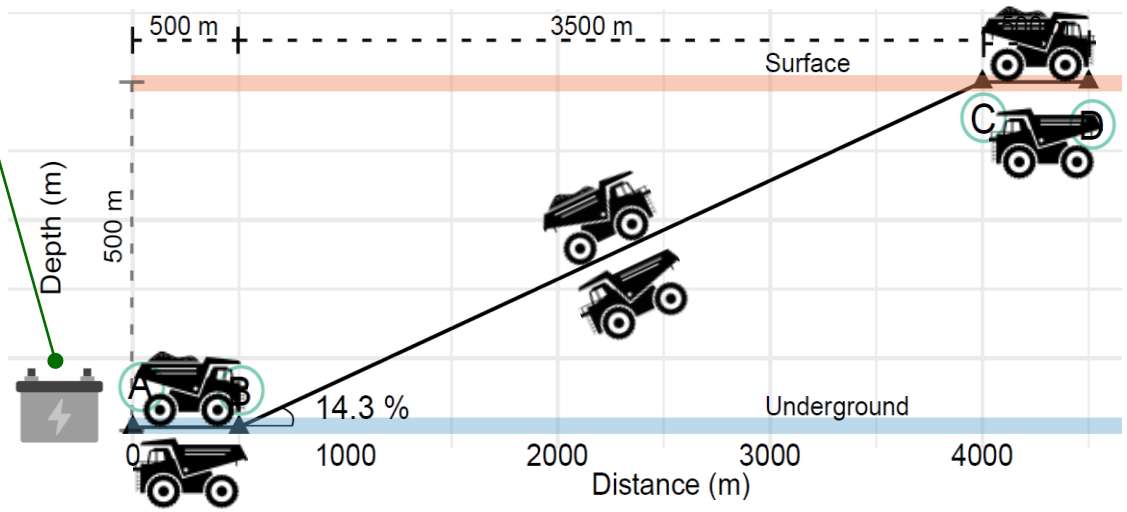

*The most recent efforts initiated by the mining industry for mining electrification.*

# 3. Problem Statement

# Problem Statement

**Charging Technology:**

- Rated voltage, power, & energy of the charging technologies
- Location of charging, swapping stations or trolly-assist charging




**Electric Truck:**

- Power and energy requirements
- Regenerative braking
- Battery onboard the truck

**Operational Parameters:**

- Battery recharging frequency
- impact of onboard battery size and charging frequency on productivity
- Truck's speed to achieve best cost vs productivity solution
- onboard battery replacement due to degradation

**Total CAPEX, OPEX, and salvage value of different electrification solutions**

# Problem Statement

Low battery energy density

Effect of battery size on energy consumption

Effect of battery C-rating on battery size

Effect of battery C-rating on charging technology size

Effect of battery size on productivity

Effect of charging rate and schedule on productivity

Effect of charging technology location

Effect of charging technology type on charging rate and battery replacement

# Problem Statement

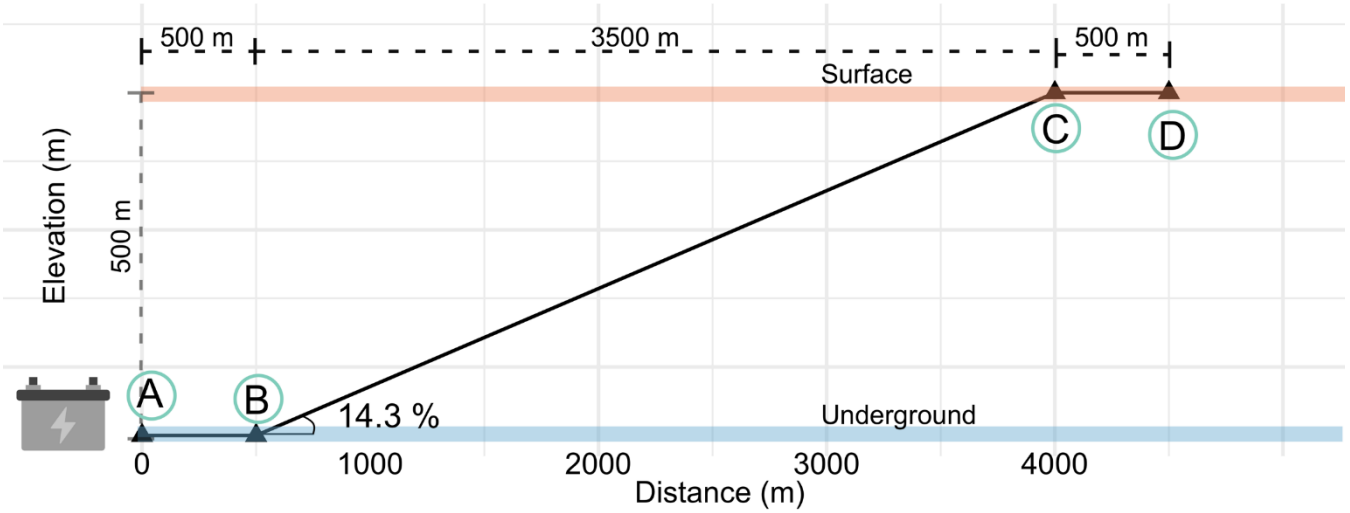
**Optimal system design includes determining the charging technology and onboard battery size**

- Minimum CAPEX and OPEX
- Maximum payload per shift
- On-board battery size

- Charging/swapping technology location in the haulage profile
- Charging technology design parameters
- Charging and fleet scheduling

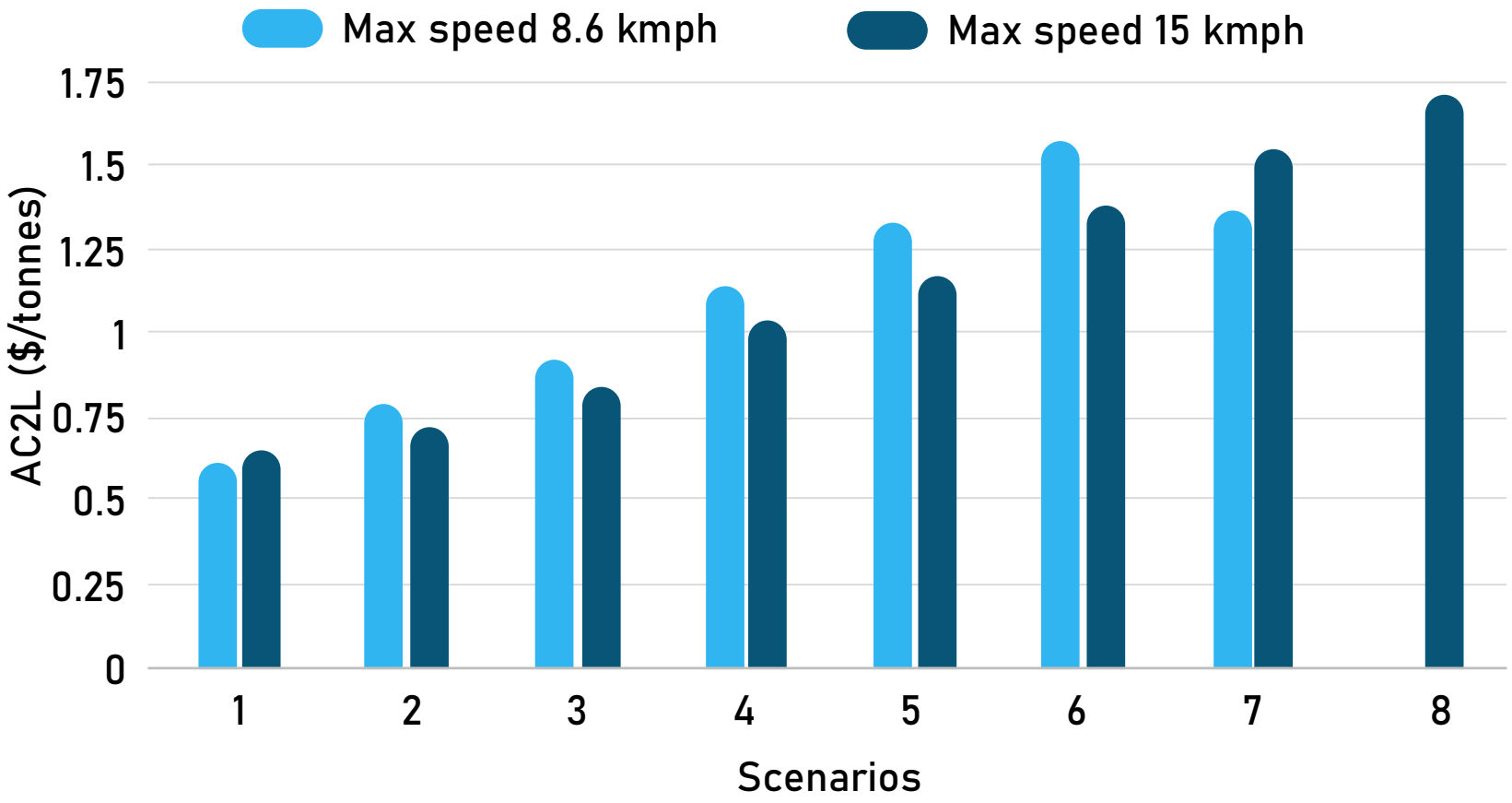
# 4. Simulation Results

# Simulation Results: Setup



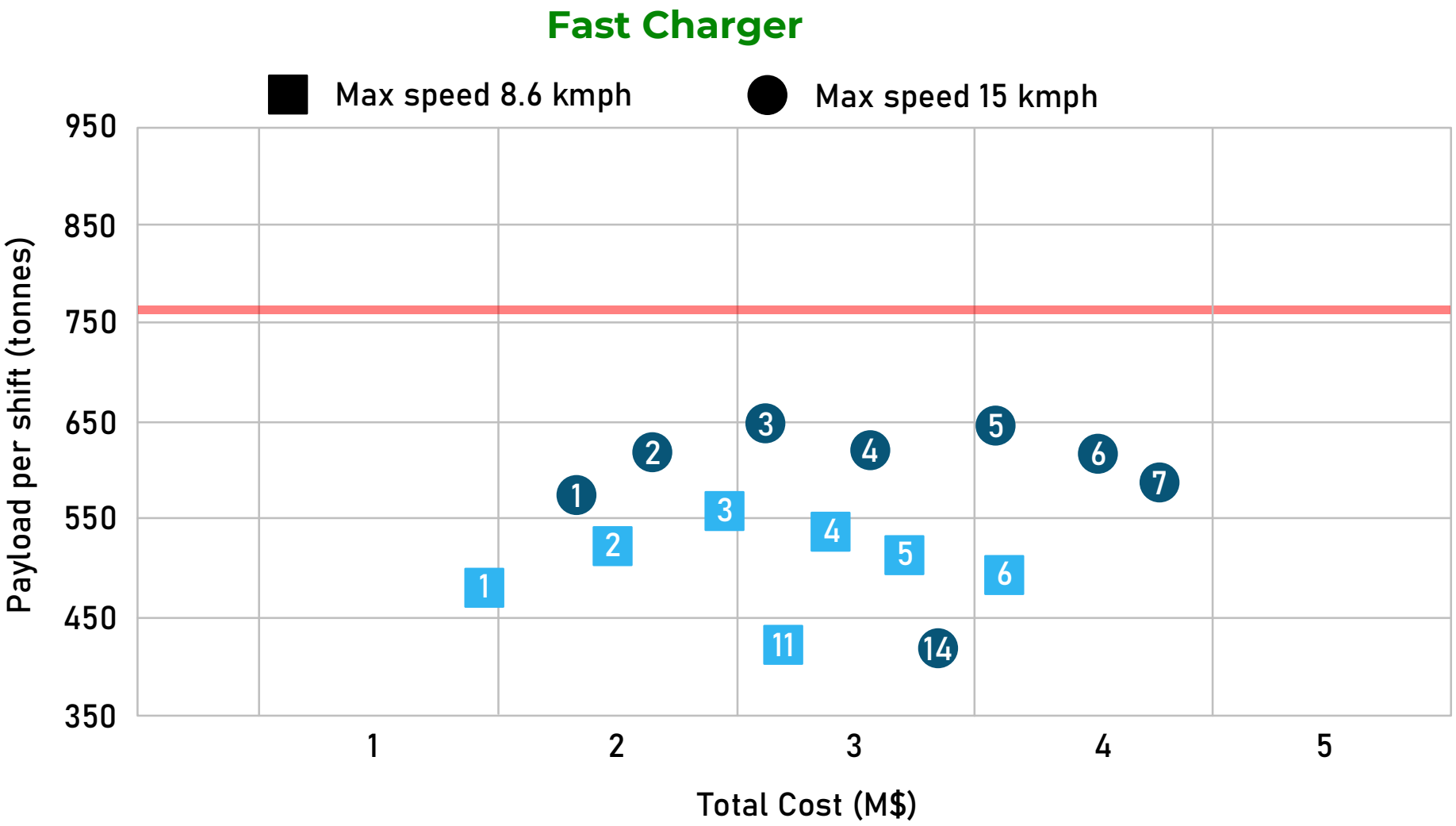

- 500m elevation change
- Truck loaded at the bottom and uploaded at the top
- 1:7 gradient (14%) is commonly used
- 7 years mine lifetime
- Evaluating the impact of charging technology, truck's speed, charging frequency –
  - 16 scenarios for fast charging and swapping
  - 10 scenarios for trolley-assist charging

# Simulation Results: Fast Charger



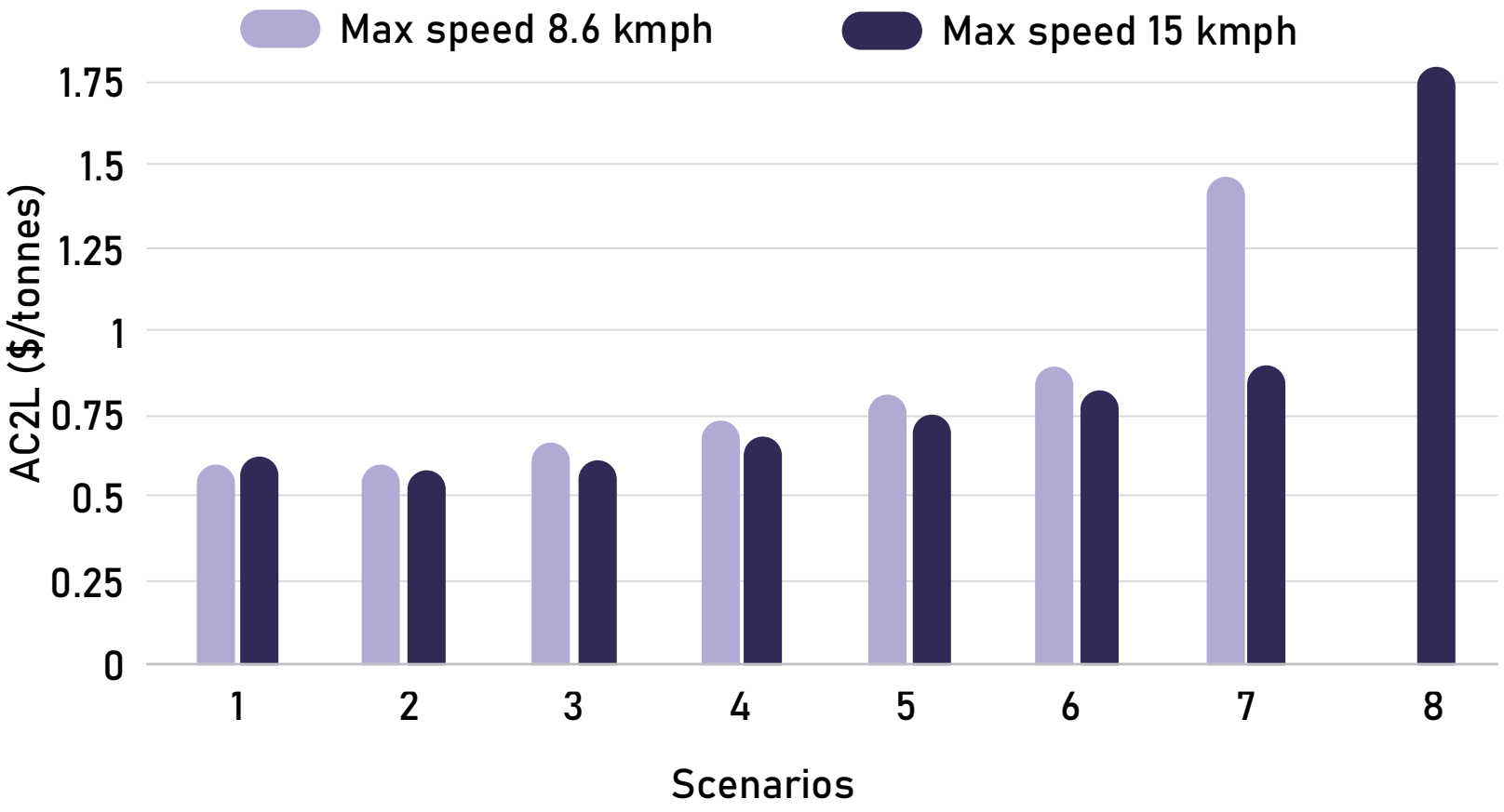

Each scenario shows a frequency of battery recharging.
For example, scenario 1 = charging at the end of each haul trip

# Simulation Results: Cost vs. Productivity Trade-Off



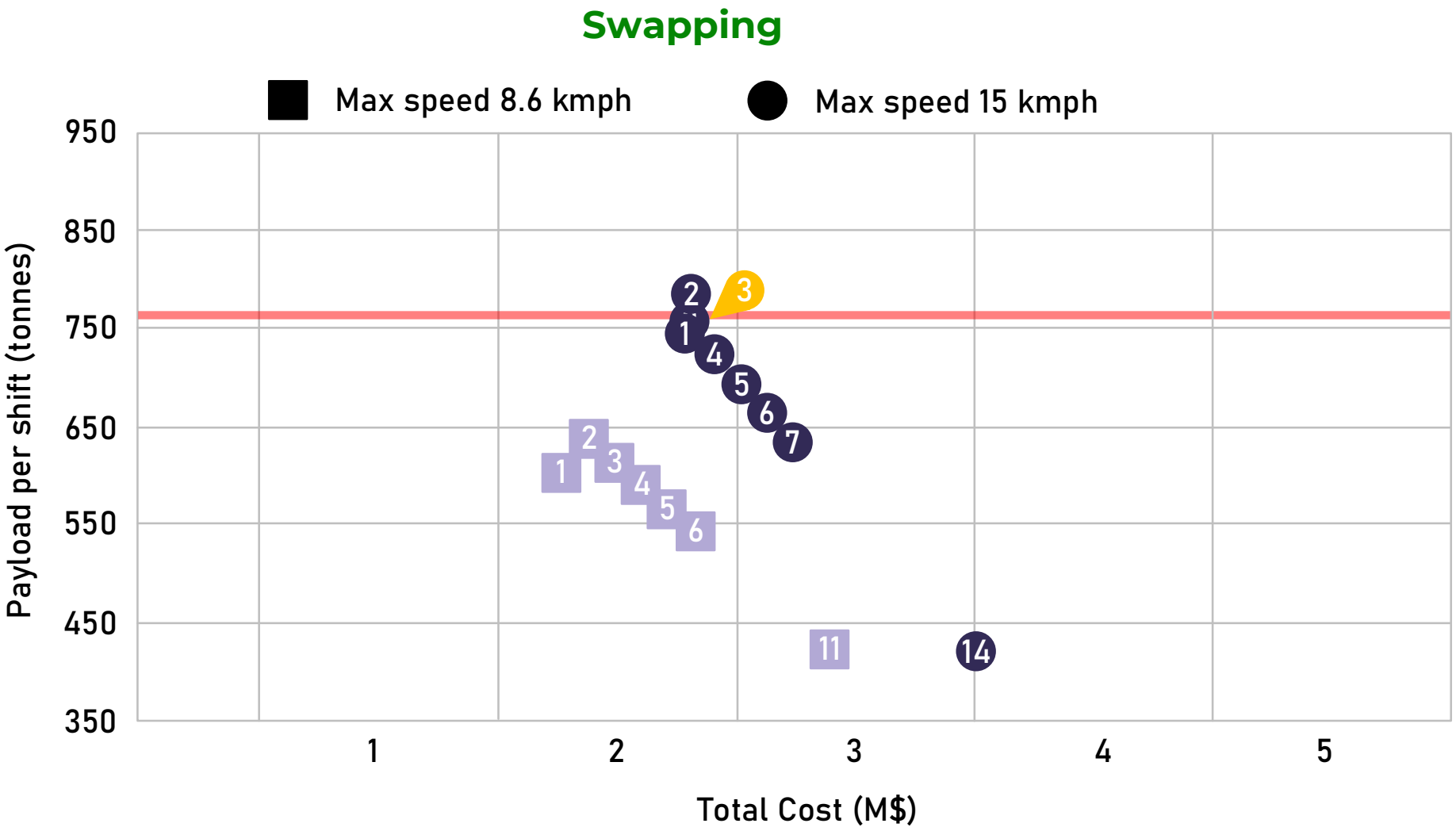

Each scenario shows a frequency of battery recharging.

# Simulation Results: Swapping



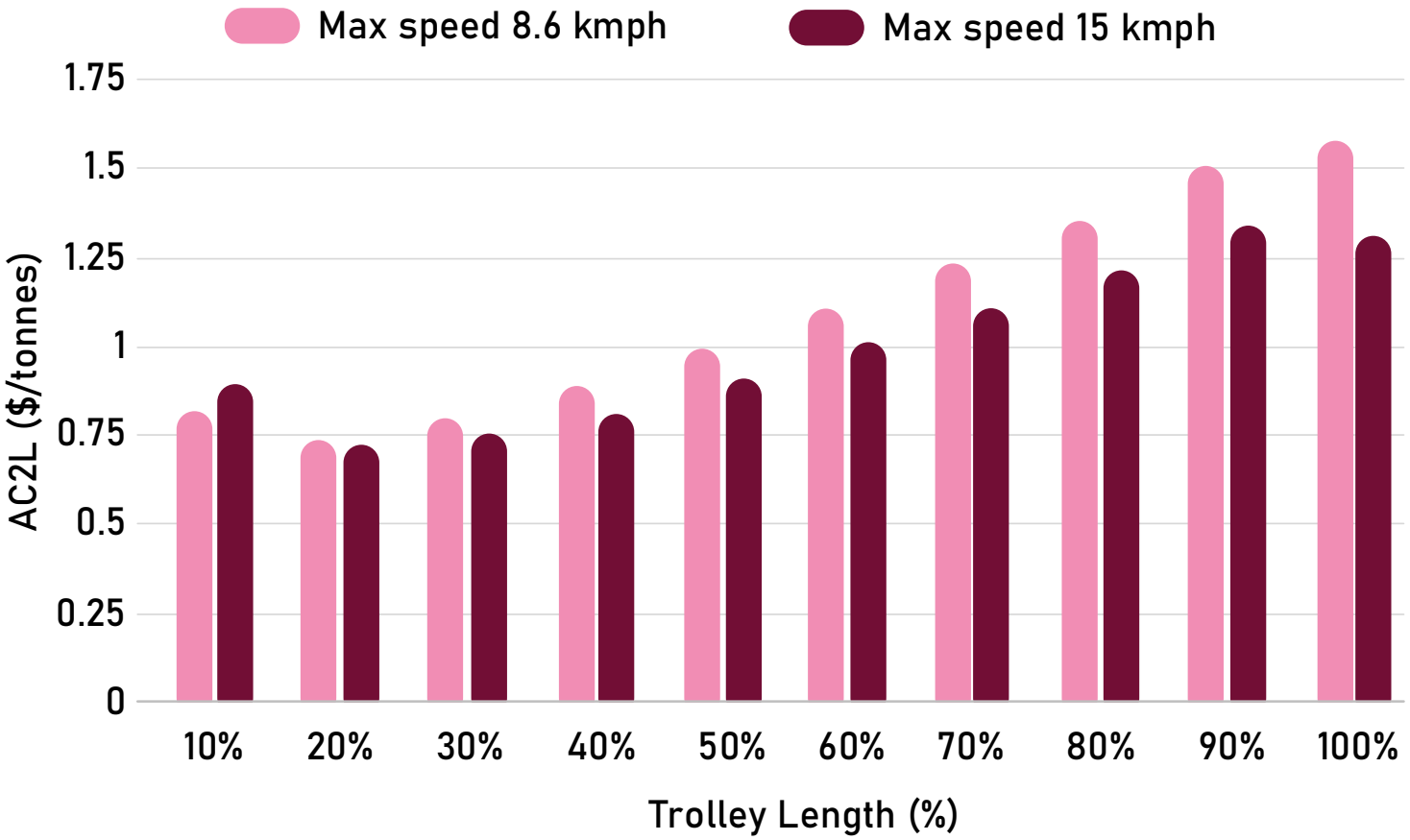

Each scenario shows a frequency of battery recharging.
For example, scenario 1 = charging at the end of each haul trip

# Simulation Results: Cost vs. Productivity Trade-Off

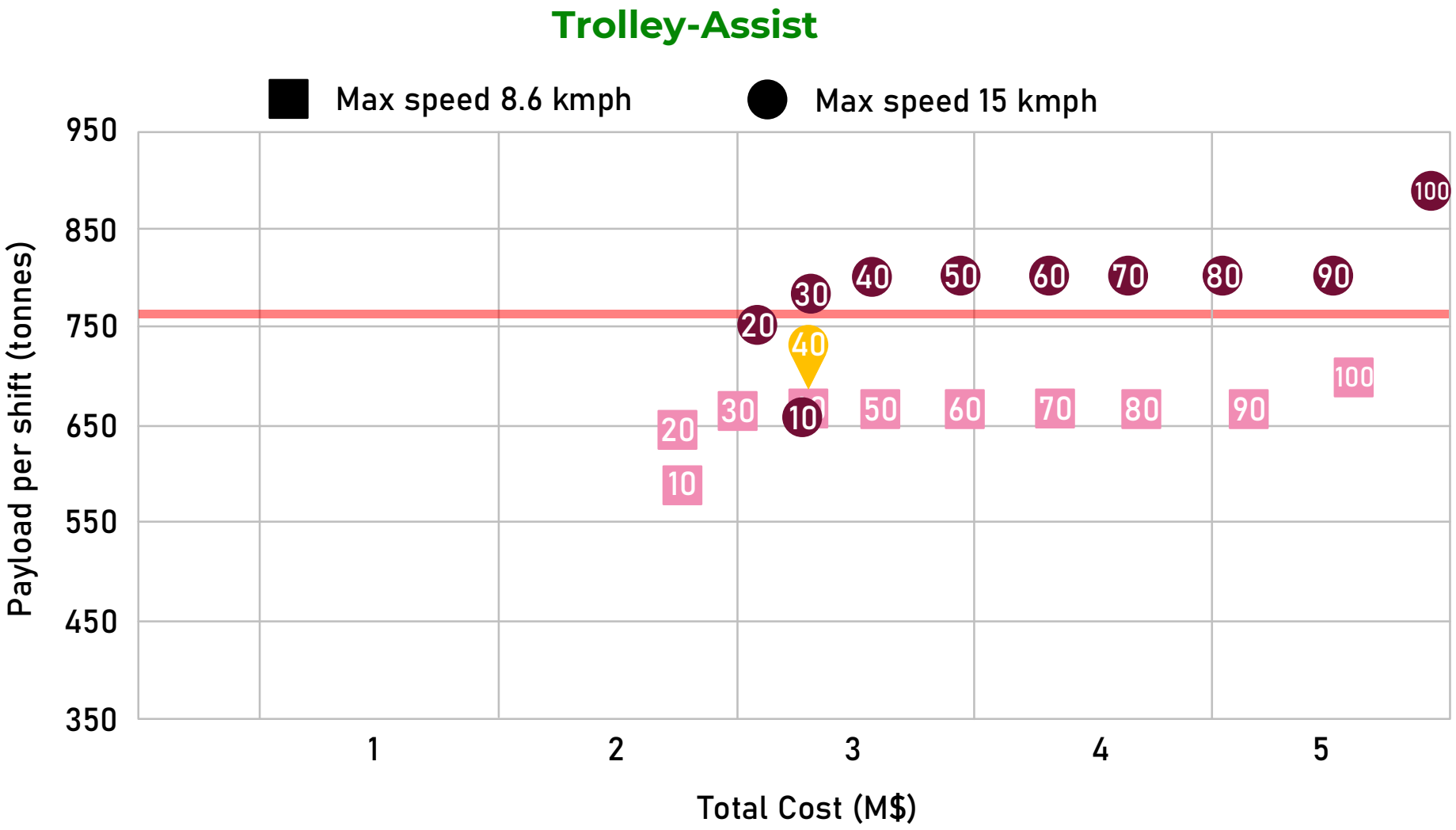



Each scenario shows a frequency of battery recharging.

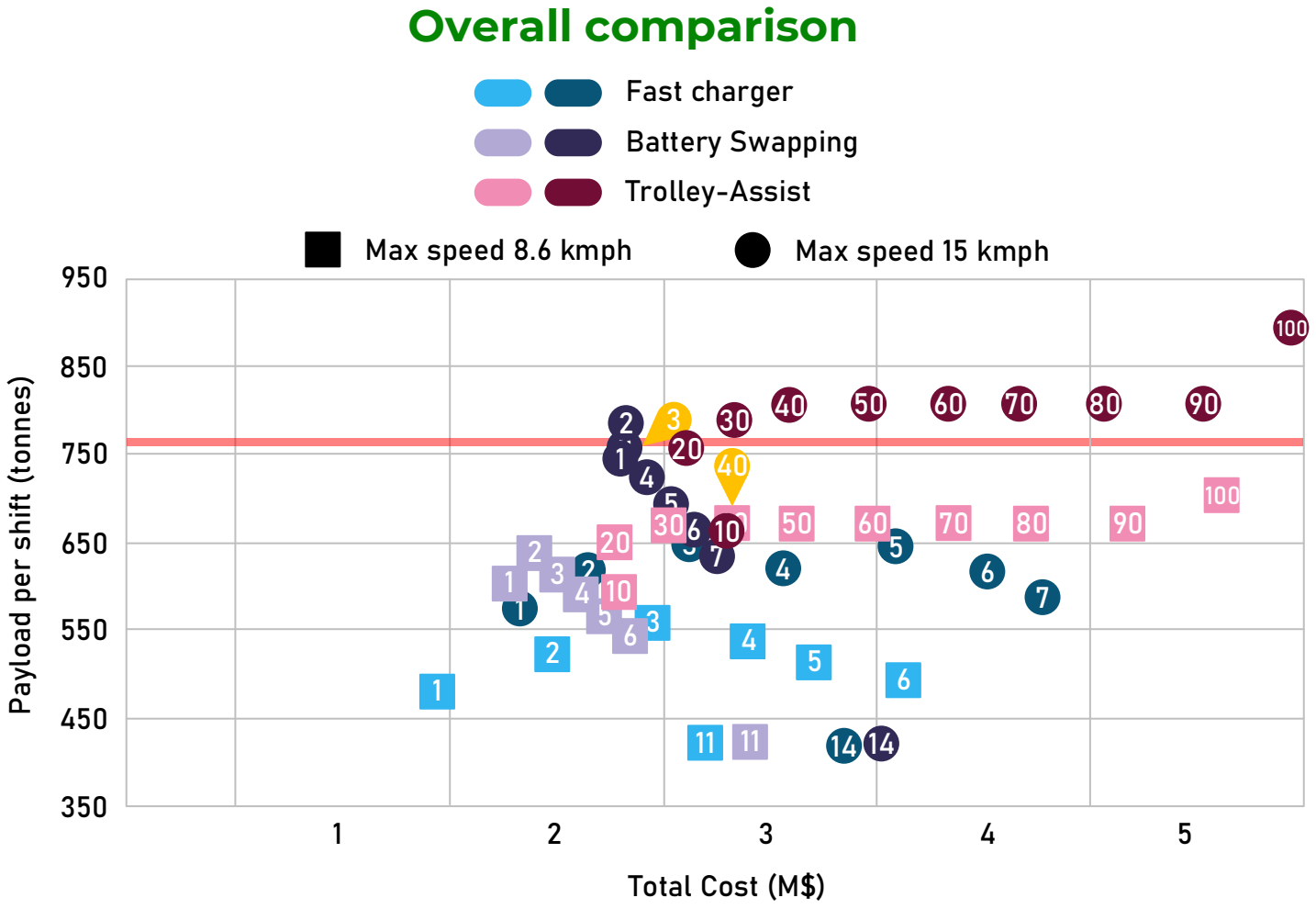
# Simulation Results: Trolley-Assist



# Simulation Results: Cost vs. Productivity Trade-Off



# Simulation Results: Cost vs. Productivity Trade-Off



# 5. Our Service

# Our Service

We offer four primary services that can assist your site towards the net zero transition:

1

### ELECTRIFICATION ASSESSMENT

We will work with you to identify the potential to electrify your specific site using extensive data analysis to deliver your custom roadmap for action.

2

### MOBILE FLEET

Armed with this information we can help guide you through the selection of appropriate technology, considering the optimal size of charging mechanism and onboard battery systems to meet your needs.

3

### FLEET MANAGEMENT SYSTEM

We can develop and customise software tools to help you manage your site's electric vehicle fleets and associated infrastructure.

4

### ENERGY INFRASTRUCTURE DESIGN

We can also design your optimal energy infrastructure to ensure that EV fleet and electric machinery operates efficiently  and reliably.

# Q&A

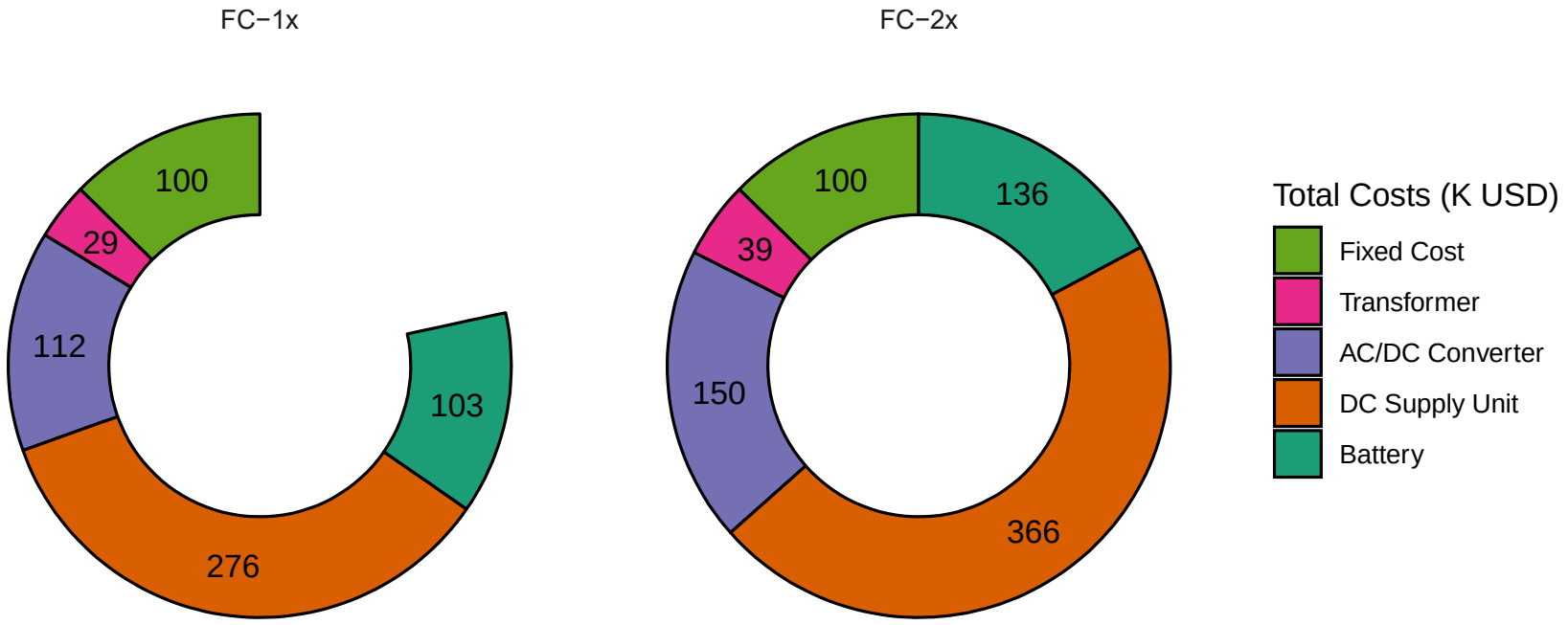
**“FROM UNDERSTANDING THE EXACT INVENTORY REQUIRED, TO CALCULATING THE COSTS OF IMPLEMENTATION, OUR TOOLS WILL HELP SUPPORT MINING COMPANIES TO MAKE A SUCCESSFUL TRANSITION TO BEVS.” — Dr Ali Pourmousavi Kani**

## Contact us

E: a.pourm@adelaide.edu.au
P: 0413 740 277

Scan for more

# Simulation Results: Fast Charger

| Trips<br>Uphill speed | Min<br>1x | Min<br>2x | Max<br>1x | Max<br>2x |
|---|---|---|---|---|
| **CASE 1: FAST CHARGING** | | | | |
| Battery charge time (hrs per trip) | 0.32 | 0.24 | - | - |
| Trip duration (hrs) | 1.1 | 0.84 | 0.78 | 0.6 |
| Trips per shift | 8 | 10 | 11 | 14 |
| Payload (tonnes/shift) | 482 | 575 | 421 | 419 |
| **Key metrics** | | | | |
| Payload (relative to diesel vehicle) | 0.7 | 0.83 | 0.61 | 0.61 |
| Power (peak, supply, MW) | 0.53 | 0.72 | 1.23 | 1.56 |
| Battery capacity (MWh) | 0.27 | 0.36 | 2.46 | 3.11 |
| Battery cycles per year (1000s) | 4.7 | 4.3 | 0.7 | 0.7 |
| Battery storage over 7 yrs (MWh) | 4.53 | 5.75 | 7.37 | 9.35 |

# Simulation Results: Fast Charger